1 Introduction

At the heart of the paper's quantification, the estimated efficiency gaps enter directly as a wedge on public spending: an efficiency gap of, say, 0.35 implies that 35 percent of each unit spent does not reach the productive public capital stock—lost to over-invoicing, poor project selection, weak implementation, or inadequate maintenance. The appeal of this formulation is that it separates the level of spending from its productivity—a country spending as much as its peers but with a larger gap wastes a higher share—so that closing the gap raises output without additional budget resources.

3 Measuring Spending Efficiency: Methodology and SACU Results

This section explains the stochastic-frontier methodology used in the Fiscal Monitor (IMF 2025a), reports the SACU-specific efficiency-gap estimates that enter the paper's simulations, and discusses interpretation and caveats.

3.1 Methodology

This paper relies on the spending-efficiency measures established in Bańkowski, Cho, and Sher (2026), which underpin the October 2025 Fiscal Monitor (IMF 2025a). Their approach compares public spending (the input) with the socioeconomic outcomes it is meant to deliver (the output) in each of the four spending sectors covered by the framework—public investment, education, health, and research and development.[1] Stochastic frontier analysis (SFA) estimates a best-practice production function from a cross-country panel and measures each country's efficiency as its distance below that frontier. Unlike non-parametric frontiers such as data envelopment analysis, SFA separates genuine inefficiency from statistical noise. It also handles persistent structural differences between countries—such as being landlocked or having a large private health sector. These are absorbed separately, through the four-component ("true random effects") estimator of Filippini and Greene (2016). The estimator distinguishes time-invariant structural heterogeneity from persistent inefficiency, so the former is not mislabelled as the latter.

Inputs are measured as a five-year moving average of real public spending per capita in purchasing-power-parity terms; outputs combine multiple quantity and quality indicators for each sector (for public investment, for example, road and electricity access alongside infrastructure-quality ratings; for education, enrolment, completion, and learning measures; for health, life expectancy, immunization, and health-system capacity). To avoid sensitivity to any single indicator, scores are averaged across all combinations of the available indicators. Because the estimator is time-varying, it delivers a separate score for each individual country in each individual year, rather than a single cross-sectional snapshot. Each score lies between 0 and 1, with 1 denoting a country on the frontier. Throughout this paper we work with the complementary *efficiency gap*—one minus the score—so that 0 denotes full efficiency and higher values denote larger shortfalls. Full methodological detail is given in Bańkowski, Cho, and Sher (2026) and the Fiscal Monitor Online Annex (IMF 2025b).

3.2 SACU Results

Efficiency gaps persist across sectors in SACU. Figure 3 reports the Fiscal Monitor efficiency gaps for the five SACU countries in three sectors—health, education, and infrastructure—at

1. R&D lies outside the scope of this paper and is not considered further; the analysis focuses on public investment, education, and health.

two points in time (2000 and 2023), alongside the unweighted averages for advanced economies, emerging markets, and sub-Saharan Africa. Three points stand out:

Infrastructure gaps have narrowed but remain wider than peers. The right panel of Figure 3 shows that all five SACU countries closed their infrastructure efficiency gaps substantially between 2000 and 2023, with Lesotho's dramatic improvement from a gap close to 0.9 to about 0.4 being the most striking. Even so, the SACU average remains above the advanced-economy average and close to the emerging-market level. The progress reflects both rising spending and strengthened investment-management frameworks, but the remaining gap points to scope for further gains.

Education gaps have been broadly stable. The middle panel shows education efficiency gaps of 0.30 to 0.45 across SACU, with limited change between 2000 and 2023 and South Africa and Namibia modestly above their 2000 levels.

Health gaps are persistent and larger than peers. The left panel shows health efficiency gaps of 0.35 to 0.50 for SACU countries, little changed since 2000, and above the peer-group averages. Health outcomes in SACU continue to reflect the lingering effects of the HIV/AIDS epidemic, but also structural features of the health systems—urban-rural coverage gaps, workforce pressures, and supply-chain weaknesses—that limit how efficiently spending translates into outcomes.

Infrastructure efficiency differences appear small but place SACU countries far apart in international comparison. Figure 4 sets the infrastructure gap in global context. It plots the 2023 infrastructure efficiency gap for every country in the Fiscal Monitor database, ordered from highest to lowest gap, with the five SACU countries highlighted. Namibia, Lesotho, and Eswatini sit close to the median of the global distribution; Botswana and South Africa are modestly below the median. None of the five SACU countries is among the best performers, confirming that the SACU distribution represents a set of economies with meaningful but not extreme scope for improvement.

Table 1 summarizes the country-specific efficiency gaps that feed the simulation exercise in Sections 5 and 6. The infrastructure gap is measured for the latest available year (2023). For human capital, we report both the education-specific gap (latest year 2024) and the health-specific gap (2023), as well as a blended gap defined as the simple average of the two. The blended gap is used in the model because the Fiscal Monitor DSGE framework treats education and health as a single public human-capital stock (schools and hospitals). The education-specific gap is reported separately because one of the reallocation variants targets education-sector investment only.

Table 1: SACU Efficiency Gaps, Latest Available Year

Efficiency gap, 0 to 1 scale; higher values indicate larger gaps. Used as inputs to the DSGE simulations.

Country	Infrastructure (e_{GI}, 2023)	Education (e_{GH}^{edu}, 2024)	Health (e_{GH}^{hlt}, 2023)	Blended HC (e_{GH}^{blend})
Botswana	0.34	0.38	0.40	0.39
Eswatini	0.39	0.35	0.41	0.38
Lesotho	0.40	0.42	0.47	0.45
Namibia	0.41	0.35	0.37	0.36
South Africa	0.33	0.37	0.37	0.37

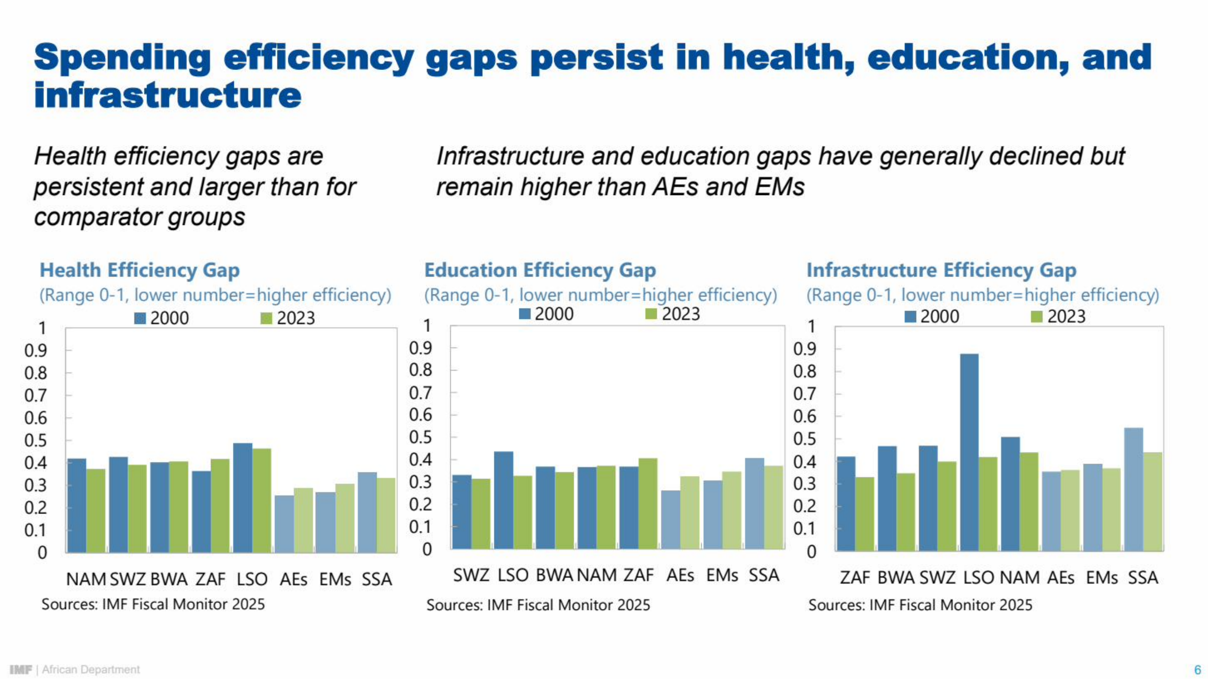
Source: IMF Fiscal Monitor (2025) efficiency-score database, extracted for the FM2025_SACU_DSGE model.
Note: Blended human-capital gap is the simple average of the education and health gaps.

3.3 Interpretation and Caveats

Three interpretive caveats apply:

Figure 3: Efficiency Gaps in Health, Education, and Infrastructure Spending

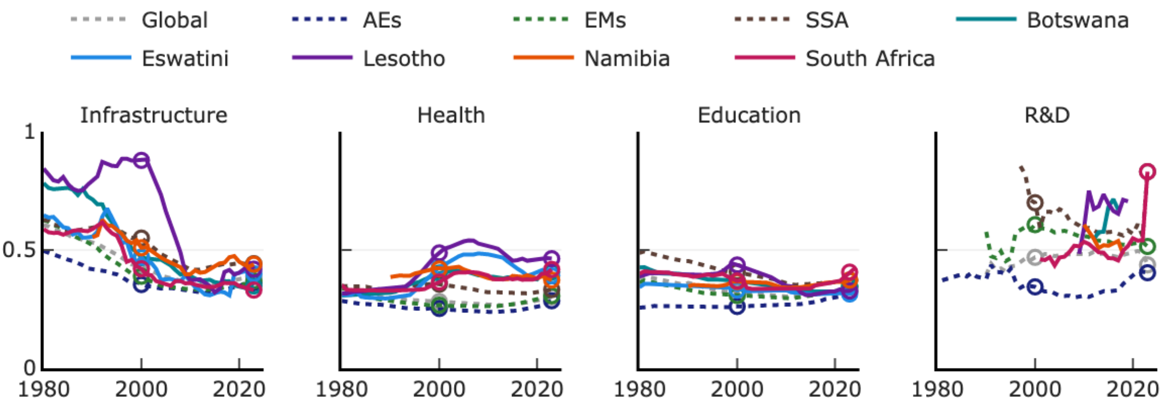
Efficiency gaps, 0 to 1 scale. Higher values indicate larger gaps.



Source: Reproduced from IMF (2026), based on IMF Fiscal Monitor (2025) efficiency-score database.

[Replication from source — for comparison with Figure 3 above]

(Efficiency gap, 0–1 scale; higher values indicate larger gaps)

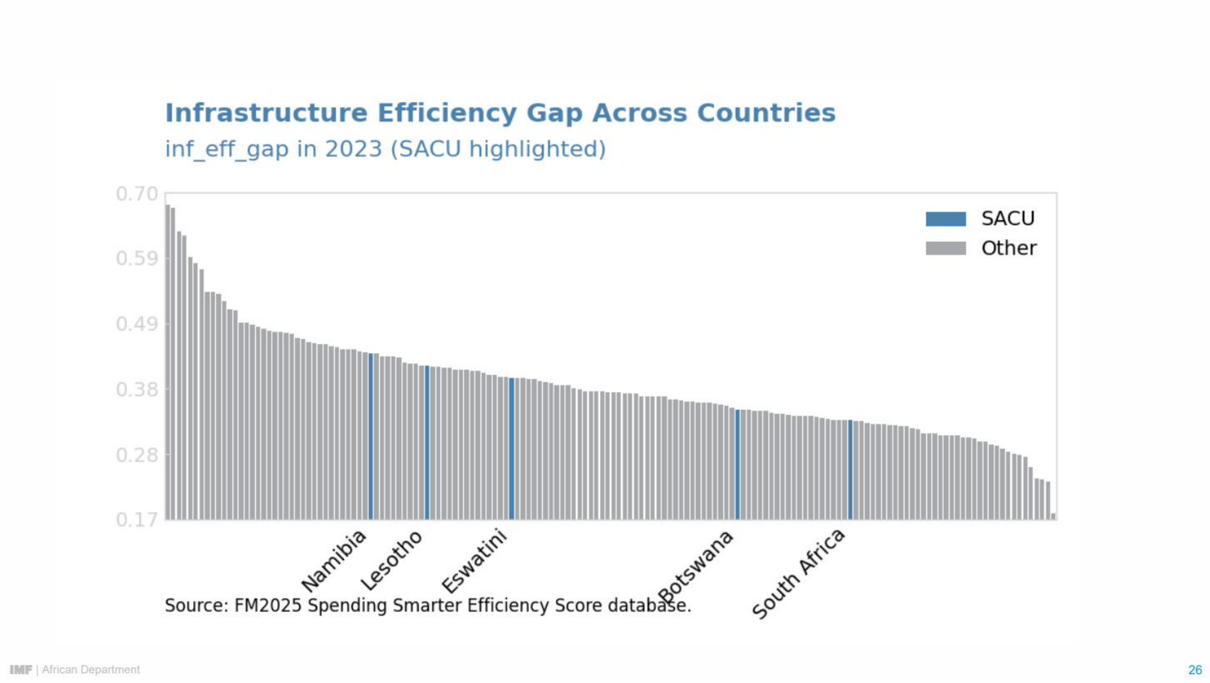


Source: IMF staff estimates.

Note: Solid lines show the five SACU countries; dotted lines show reference averages—global, advanced economies (AEs), emerging markets (EMs), and sub-Saharan Africa (SSA). Hollow circles mark 2000 and 2023.

Figure 4: Infrastructure Efficiency Gap Across Countries, 2023

Cross-country distribution of infrastructure efficiency gap, with SACU countries highlighted.



Source: Reproduced from IMF (2026).
Note: Each bar is one country; SACU members shown in blue.

First, efficiency gaps conflate several underlying factors—institutional capacity, governance quality, the project mix, and country-specific shocks—and do not, on their own, identify the mechanism by which resources are lost. Section 4 links the measured gaps to institutional evidence from the Public Investment Management Assessments to address this.

Second, estimates are sensitive to the frontier specification. The Fiscal Monitor Online Annex (IMF 2025b) reports that alternative weightings of the underlying efficiency-score models can shift the level of estimated gaps by up to 26 percentage points in the extreme case. We use the baseline specification throughout, with sensitivity analysis discussed in Section 6.

Third, there are data-window limitations. The SFA uses a five-year moving average, which means the 2023 estimate reflects spending and outcomes between 2019 and 2023—a period disrupted by the COVID-19 shock, the subsequent SACU revenue boom, and commodity-price swings that affected Botswana in particular. Estimates should therefore be interpreted as reflecting the medium-run institutional capacity of each system rather than a precise snapshot of any single year.

References

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